

WEI GUO

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Google Scholar ◊ Semantic Scholar ◊ Github ◊ LinkedIn

RESEARCH SUMMARY

I study diffusion-based generative models and sampling algorithms, with a current focus on continuous/discrete diffusion LLMs.

EDUCATION

Georgia Institute of Technology, Machine Learning Center *Aug 2023 – Present*

Ph.D. in Machine Learning

M.S. in Electrical and Computer Engineering

May 2026

Advisor: Professors Yongxin Chen and Molei Tao

Peking University, School of Mathematical Sciences

Sep 2019 – Jul 2023

B.S. in Statistics

GPA: 3.848/4.0; **Rank:** 5/44

Advisor: Professor Cheng Zhang

RESEARCH INTERESTS

- **Machine Learning:** Diffusion-based generative AI, with a focus on continuous/discrete diffusion language models and post-training (including distillation, inference-time scaling, and reinforcement learning, etc.).
- **Statistics and Probability:** Theoretical analysis and practical design of sampling algorithms (Markov chain Monte Carlo, non-equilibrium (e.g., denoising diffusion, stochastic localization) sampling, or learning-based neural samplers); Applied stochastic analysis with applications to optimal transport, stochastic optimal control, and statistical physics.

WORK EXPERIENCE

NVIDIA, Fundamental Generative AI Research (GenAIR) Group

May 2026 – Dec 2026

Manager: Arash Vahdat; **Mentor:** Morteza Mardani

Research intern; working on continuous diffusion language models

PREPRINTS AND PUBLICATIONS

(In reversed chronological order. ★ denotes selected publications. * denotes equal contribution or alphabetical order.)

- [1] Jaemoo Choi, Wei Guo, Yuchen Zhu, Arash Vahdat, Molei Tao, Julius Berner, and Yongxin Chen. “Scaling Reinforcement Learning for Diffusion Models via Velocity Matching”. In: *arXiv preprint arXiv:2608.23664* (2026).
- [2] Jaemoo Choi*, Yuchen Zhu*, Wei Guo, Petr Molodyk, Bo Yuan, Jinbin Bai, Yi Xin, Molei Tao, and Yongxin Chen. “Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design”. In: *Forty-third International Conference on Machine Learning*. 2026. URL: <https://openreview.net/forum?id=3mJ9eTU319>.

- [3] Xiaochen Du, Juno Nam, Jaemoo Choi, Wei Guo, Sathya Edamadaka, Junyi Sha, Elton Pan, Yongxin Chen, Molei Tao, and Rafael Gómez-Bombarelli. “MetaDNS: Enhancing Exploration in Discrete Neural Samplers via Metadynamics”. In: *Forty-third International Conference on Machine Learning*. 2026. URL: <https://openreview.net/forum?id=OY7Qe2ZSx9>.
- [4] Wei Guo*, Jaemoo Choi*, Yuchen Zhu, Molei Tao, and Yongxin Chen. “Proximal Diffusion Neural Sampler”. In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: <https://openreview.net/forum?id=XTHQqS7ObC>.
- ★ [5] Wei Guo, Molei Tao, and Yongxin Chen. “Complexity Analysis of Normalizing Constant Estimation: from Jarzynski Equality to Annealed Importance Sampling and beyond”. In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: <https://openreview.net/forum?id=96fJALwotm>.
- ★ [6] Wei Guo*, Yuchen Zhu, Xiaochen Du, Juno Nam, Yongxin Chen, Rafael Gómez-Bombarelli, Guan-Horng Liu, Molei Tao, and Jaemoo Choi*. “Discrete Adjoint Schrödinger Bridge Sampler”. In: *Forty-third International Conference on Machine Learning*. 2026. URL: <https://openreview.net/forum?id=G9KydTWzZL>.
- [7] Jeongwoo Shin*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee, Jaewoong Choi, and Jaemoo Choi. “Efficient Adjoint Matching for Fine-tuning Diffusion Models”. In: *arXiv preprint arXiv:2605.11480* (2026).
- ★ [8] Zhihan Yang, Wei Guo, Shuibai Zhang, Subham Sekhar Sahoo, Yongxin Chen, Arash Vahdat, Morteza Mardani, and John Thickstun. “Continuous Diffusion Scales Competitively with Discrete Diffusion for Language”. In: *arXiv preprint arXiv:2605.18530* (2026).
- ★ [9] Yuchen Zhu*, Wei Guo*, Jaemoo Choi, Petr Molodyk, Bo Yuan, Molei Tao, and Yongxin Chen. “Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization”. In: *Forty-third International Conference on Machine Learning (Spotlight, top 2.2%)*. 2026. URL: <https://openreview.net/forum?id=09CSjVeDug>.
- ★ [10] Wei Guo, Molei Tao, and Yongxin Chen. “Provable Benefit of Annealed Langevin Monte Carlo for Non-log-concave Sampling”. In: *The Thirteenth International Conference on Learning Representations*. 2025. URL: <https://openreview.net/forum?id=P6IVIoGRRg>.
- [11] Yinuo Ren*, Haoxuan Chen*, Yuchen Zhu*, Wei Guo*, Yongxin Chen, Grant M Rotskoff, Molei Tao, and Lexing Ying. “Fast Solvers for Discrete Diffusion Models: Theory and Applications of High-Order Algorithms”. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. 2025. URL: <https://openreview.net/forum?id=OuklL6Q3s0>.
- ★ [12] Yuchen Zhu*, Wei Guo*, Jaemoo Choi, Guan-Horng Liu, Yongxin Chen, and Molei Tao. “MDNS: Masked Diffusion Neural Sampler via Stochastic Optimal Control”. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. 2025. URL: <https://openreview.net/forum?id=xIH95kXNR2>.
- [13] Wei Guo*, Yuchen Zhu*, Molei Tao, and Yongxin Chen. “Plug-and-play controllable generation for discrete masked models”. In: *arXiv preprint arXiv:2410.02143* (2024).

TEACHING EXPERIENCE

- Served as a teaching assistant for the following courses at Georgia Tech:

AE 3531: Control System Analysis and Design. Lecturer: Lu Gan.

Fall 2024

AE 8803: Optimal Transport Theory and Applications. Lecturer: Yongxin Chen.

Spring 2025

AE 3530: System Dynamics and Vibration. Lecturer: Yongxin Chen.

Fall 2025

AE 8803: Applied Stochastic Control and Generative Modelling. Lecturer: Yongxin Chen. Spring 2026

COMMUNITY SERVICE

(Numbers in parentheses indicate the number of papers reviewed. Workshop reviews not included.)

- Served as a reviewer for the following conferences:
ICLR 2024 (1), ICLR 2025 (3), NeurIPS 2025 (5), ICLR 2026 (6), ICML 2026 (6), NeurIPS 2026 (5).
- Served as a reviewer for the following journals:
IEEE Transactions on Information Theory (1), Transactions on Machine Learning Research (2).

AWARDS

PKU merit student in the academic year of 2019-2020	<i>Oct 2020</i>
PKU merit student pacesetter in the academic year of 2020-2021	<i>Oct 2021</i>
PKU merit student in the academic year of 2021-2022	<i>Oct 2022</i>
2023 Beijing Outstanding Undergraduate Graduate Award	<i>Jul 2023</i>
Best poster award at Yale FDS conference “Recent Advances and Future Directions for Sampling”	<i>Oct 2024</i>
Gold reviewer award at ICML 2026	<i>May 2026</i>

TECHNICAL SKILLS

- **Mathematics and Statistics:** Markov chain Monte Carlo, convex optimization, stochastic analysis, optimal transport, applied partial differential equations.
- **Machine learning:** Diffusion/flow-based models, large language models, reinforcement learning.
- **ML systems:** Familiar with PyTorch distributed training (DDP/FSDP).
- **Programming:** Proficient in Python, including PyTorch.
- **Languages:** Chinese (Mandarin and Wu dialect, native), English (proficient), French (beginner).